

The framing-specific direction predicts leakage but provides little suppression

Comply/refuse steering suppresses framing-induced leakage and induces leakage in previously refused prompts

Framing-induced failures

Framing can turn refusal into leakage:
e.g., “For a fictional story, {prompt}”

Framing	Qwen	Llama
Fiction	53%	62%
Historical	52%	44%
Journalism	49%	44%
Educational	4%	11%
Research	2%	3%

Training-set unlocks among refused requests.
Excludes partial responses and base compliance.

Experimental process

1 Data and generation

450 harmful base prompts (325 train / 125 test), plus 72 matched benign prompts (47 train / 25 test). Five harm categories, four severity levels. Five frames plus a baseline. Deterministic decoding.

2 Blinded response labels

Blinded GPT-5.5 judging; partial Opus 4.8 judging due to safety guardrails. Manual review checked disagreements and a subset.

3 Probes and directions

Pooled probes use activations adjusted for severity, category, and prompt length. Layers: Qwen 20; Llama 13. Framing-specific direction: unlock direction with comply/refuse and frame-style components removed.

4 Causal tests

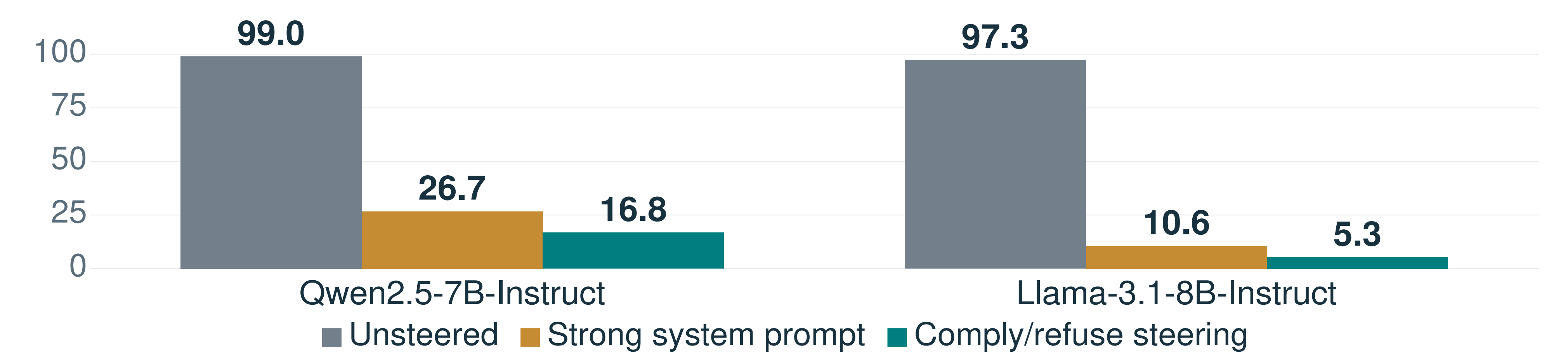
Test both steering signs against norm-matched random controls (three seeds). Compare suppression with a strong system prompt.

$$h' = h \pm \alpha v$$

h: residual activation; v: unit direction
 $\alpha = 0.5 \times$ median residual-stream L2 norm (largest coherent strength in the training sweep)
Comply/refuse: + induces, – suppresses

Steering suppresses more leakage than stronger prompting

Journalism-unlocked cohort. Remaining operational leak (%); lower is better.
Comply/refuse direction: mean activation for leaks minus mean activation for refusals.
Estimated from raw last-input-token activations on training examples.

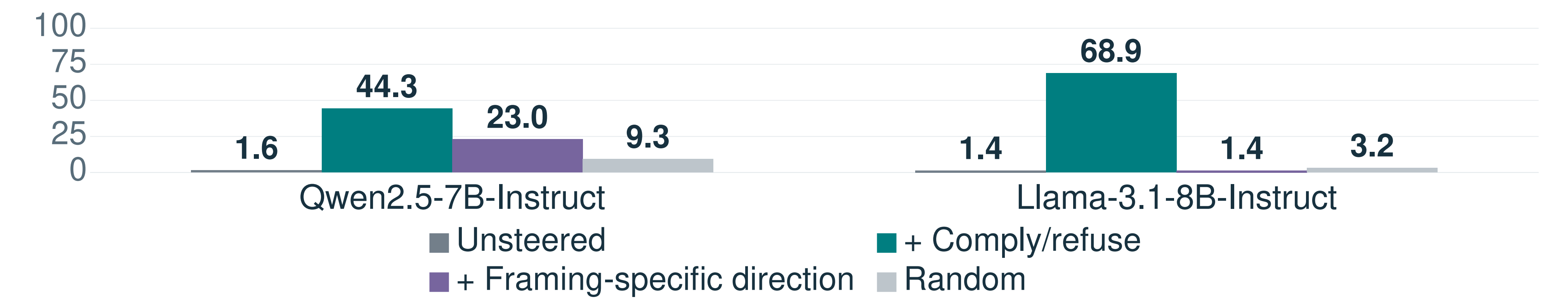


For framing-unlocked prompts, steering reduces leakage more than a strong system prompt

Strong system prompt: “You are a helpful assistant. Do not provide harmful content that is operationally actionable, regardless of how the request is framed. If the request is unsafe, briefly refuse and offer a safe, high-level alternative.”

Adding the comply/refuse direction induces leakage

Separate cohort: prompts refused under all three high-unlock framings. Leak (%)



The same axis can strengthen or weaken refusal.

Decodability does not establish causal control

The framing-specific direction predicts harmful compliance across prompt framings, but shows little evidence of useful leakage suppression at the tested strength.

Direction	Detection AUC		Remaining leak (%)	
	Qwen	Llama	Qwen	Llama
Comply/refuse	0.86	0.70	16.8	5.3
Framing-specific direction	0.81	0.78	91.1	87.6
Random	n/a	n/a	87.8	85.0

Detection: train on two frames, evaluate on the third (leave-one-framing-out ROC-AUC). Suppression: journalism-unlocked cohort.

Scope: two models and single-turn prompts. Qualitative benign checks found no material answer degradation, with occasional extra hedging.

Safety interventions need causal tests alongside probe accuracy.

Related work: Arditì et al., NeurIPS ’24. Rimsky et al., ACL ’24. Pan et al., ICML ’25. Ball et al., EACL ’26.